NFCAMPUS

A DIGITAL STUDENT IDENTIFICATION & ACCESS SYSTEM USING NFC TECHNOLOGY FOR ANDROID

PENINSULA COLLEGE GEORGETOWN DK266-03(P)

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BACKGROUND & PROBLEM STATEMENT

Our college uses RFID access cards for entry into campus buildings, libraries, and labs.

PROBLEM

While effective, this system has a major flaw: "It relies solely on a physical card".

- Students are frequently locked out due to forgotten, lost, or damaged cards
- Leading to missed classes, exams, and significant frustration.

VS

SOLUTION

The core issue is that the system identifies the card, not the person.

- Our solution leverages NFC technology on smartphones
- Creating a digital credential that integrates access control into a device students always carry.

OBJECTIVES

Develop a secure and reliable smartphone-based authentication system that supplements the existing college access card system.

SMART Criteria:

Specific	Develop a functional mobile application prototype that can emulate the college access card's credentials through NFC
Measurable	Prototype should successfully interact with test reader and trigger the "unlock" signal
Achievable	Use Android Studio and Host Card Emulation (HCE) for Android phones
Relevant	Provides a practical, modern supplement to the existing card system
Time-Bound	Project deadline: 28th March 2026

KEY FEATURES

- **User Authentication:** Email and password login system
- **OCR Integration:** Automatic card scanning
- **NFC UID Binding:** Secure card-to-account linking
- **HCE Technology:** Smartphone card emulation
- **Cloud Backend:** Firebase-powered data management
- **Secure Storage:** Encrypted credential protection

WHY NFCAMPUS

- **Convenience:** Students always carry smartphones
- **Safe Testing:** Uses a custom NFC reader prototype
- **Security:** Encrypted NFC authentication
- **Reliable Backup Access:** Works alongside existing campus cards
- **Future-Ready:** Supports smarter campus access systems

IMPLEMENTATION PLAN

Module 1: Basic Foundations

- Develop a backend server, database, and basic API endpoints
- Build an Android app with a login system, user interface and navigation

Module 2: Core Features

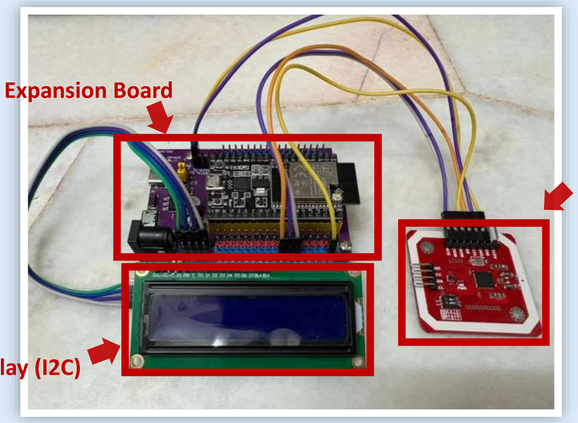
- OCR registration flow, NFC UID capture
- Password and email login, one-card-per-account system

Module 3: HCE Integration

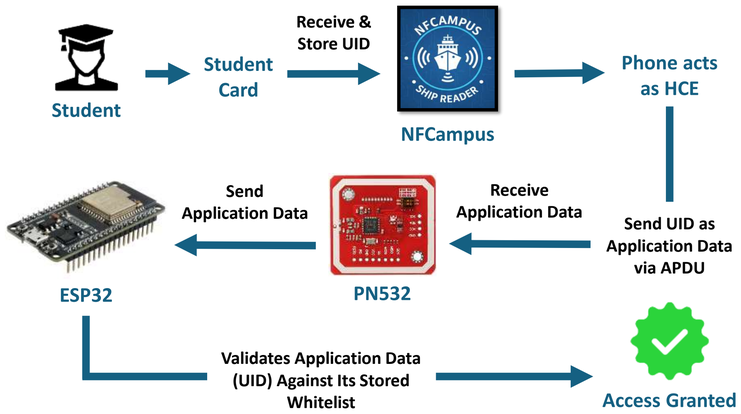
- Implement HCE service to emulate card's UID
- Integrate "Simulate Access" feature

Module 4: Testing & Polishing

- End-to-end testing, security auditing
- User testing, documentation



HOW THE SYSTEM WORKS

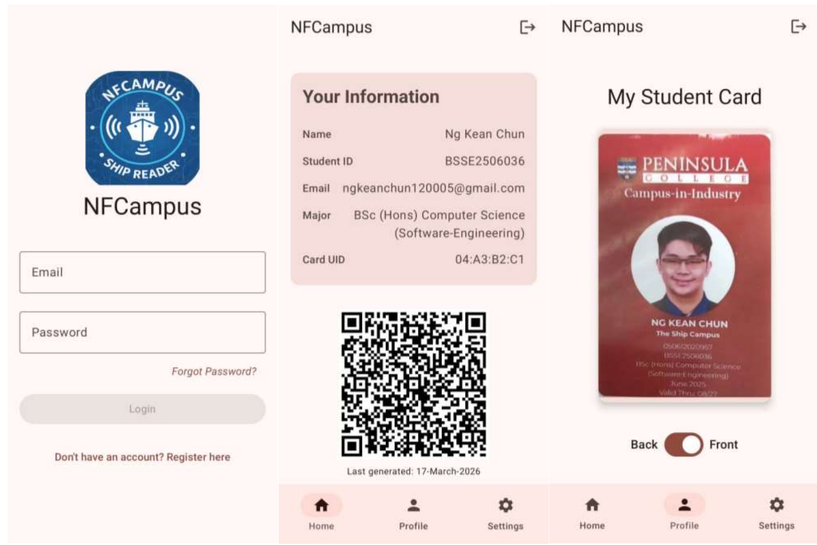


METHODOLOGY

Framework:
Agile-Scrum approach with iterative development

- **Sprints** — Two-week development cycles
- **Sprint Planning** — Persistent project tracking
- **Daily Stand-ups** — Synchronize activities and identify blockers
- **Sprint Review** — Demonstrate completed functionality
- **Retrospective** — Process improvement and refinement

SYSTEM'S GUI EXAMPLE



Login Screen | Home Screen | Profile Screen

EXPECTED OUTCOME & SIGNIFICANCE

Anticipated Results:

- A fully functional Android application
- Seamless NFC-based access control
- Enhanced security model

Contributions:

Academic: Advancement in mobile computing and security
Industrial: Modernization of campus infrastructure
Societal: Enhanced user convenience and inclusivity
Beneficiaries: Students and the academic institution

Android Studio | Firebase | Cloud Firestore | Kotlin | Node.js | Javascript | GitHub | Figma

SCAN HERE FOR REFRENCES